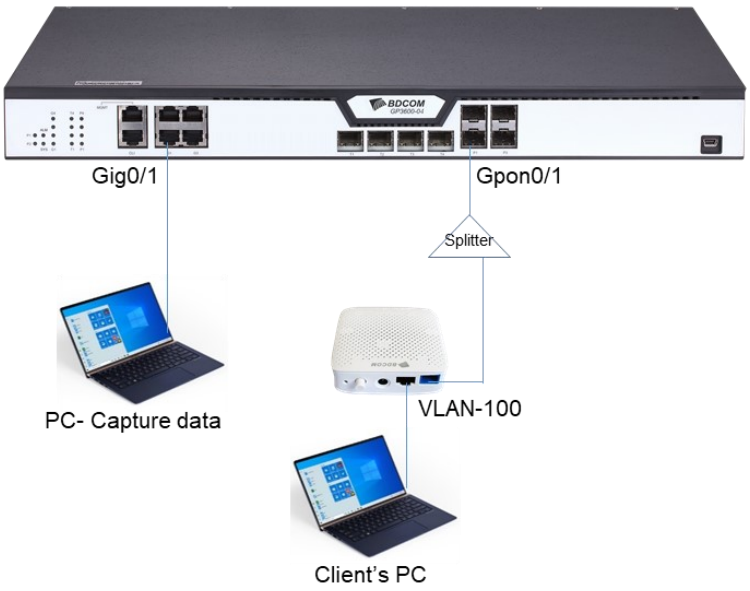
The 3 QinQ Model in GPON OLT

Topology



1. Without QinQ, do a VLAN on the LAN port of ONU

```
gpon profile onu-vlan V-100 id 4
 gpon-profile vlan mode tag
 gpon-profile vlan pvid 100 0
!

interface GPON0/1
 gpon bind-onu sn 4244434DF79D62FC 1
 switchport mode trunk
 switchport protected 1
!

interface GPON0/1:1
 gpon onu equipment-id 0000
 gpon onu tcont-virtual-port-bind-profile tvbind-default
 gpon onu flow-mapping-profile flow-mapping-default
 gpon onu virtual-port 1 gem-port 257
 gpon onu uni 1 vlan-profile V-100
```

```
> Frame 145: 60 bytes on wire (480 bits), 60 bytes captured (480 bits) on interface 0
> Ethernet II, Src: Shanghai_be:63:c8 (00:e0:0f:be:63:c8), Dst: Broadcast (ff:ff:ff:ff:ff:f
> 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 100
> Address Resolution Protocol (reply/gratuitous ARP)
```

The result: Only capture one VLAN from ONU side.

2. Simple QinQ

```
interface GigaEthernet0/1
 switchport mode dot1q-tunnel-uplink
!
interface GPON0/1
 gpon bind-onu sn 4244434DF79D62FC 1
switchport mode dot1q-translating-tunnel
 switchport pvid 200
 switchport protected 1
!
interface GPON0/1:1
 gpon onu equipment-id 0000
 gpon onu tcont-virtual-port-bind-profile tvbind-default
 gpon onu flow-mapping-profile flow-mapping-default
 gpon onu virtual-port 1 gem-port 257
 gpon onu uni 1 vlan-profile V-100
```

```
> Frame 549: 68 bytes on wire (544 bits), 68 bytes captured (544 bits) on interface 0
> Ethernet II, Src: SamsungE_bb:4b:c2 (e8:03:9a:bb:4b:c2), Dst: SpeedDra_03:02:e6 (00:13:3b:03:02:e6)
> 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 200
> 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 100
> Address Resolution Protocol (request)
```

The result: Capture 2 VLAN, inner VLAN 100, outer VLAN 200

3. QinQ

```
interface GigaEthernet0/1

 switchport mode dot1q-tunnel-uplink

!

interface GPON0/1

 gpon bind-onu sn 4244434DF79D62FC 1
 switchport mode dot1q-translating-tunnel
 switchport protected 1
!

interface GPON0/1:1

 gpon onu equipment-id 0000
 gpon onu tcont-virtual-port-bind-profile tvbind-default
 gpon onu flow-mapping-profile flow-mapping-default
 gpon onu virtual-port 1 gem-port 257
 gpon onu virtual-port 1 dot1q-translating-tunnel mode QinQ translate 100 200 0
 gpon onu uni 1 vlan-profile V-100
```

```
> Ethernet II, Src: SpeedDra_03:02:e6 (00:13:3b:03:02:e6), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 200
> 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 100
```

The result: Capture 2 VLAN, outer VLAN 200, inner VLAN 100

4. Flat QinQ

```
interface GigaEthernet0/1

 switchport mode dot1q-tunnel-uplink

!

interface GPON0/1
 gpon bind-onu sn 4244434DF79D62FC 1
 switchport mode dot1q-translating-tunnel
 switchport protected 1
!

interface GPON0/1:1
 gpon onu equipment-id 0000
 gpon onu tcont-virtual-port-bind-profile tvbind-default
 gpon onu flow-mapping-profile flow-mapping-default
```

gpon onu virtual-port 1 gem-port 257

gpon onu virtual-port 1 dot1q-translating-tunnel mode flat translate 1to1 100 200

gpon onu uni 1 vlan-profile ***V-100***

```
> Frame 7: 64 bytes on wire (512 bits), 64 bytes captured (512 bits) on interface 0
> Ethernet II, Src: SpeedDra_03:02:e6 (00:13:3b:03:02:e6), Dst: Broadcast (ff:ff:ff:ff:ff:ff)
> 802.1Q Virtual LAN, PRI: 0, CFI: 0, ID: 200
> Address Resolution Protocol (request)
```

The result: Only capture outer VLAN 200